THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–10) | Class 6 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 81	First term + (12-1) x 7 = 4+11x7
2	**(c)** 240	The two angles between the hands add up to 360°.
3	**(d)** 1234	End with the smallest even digit, then arrange the rest in increasing order.
4	**(d)** 100	Add the two biggest numbers.
5	**(a)** 10	$48 = 2^4 \times 3$; use $(e_1+1)(e_2+1)\ldots$
6	**(c)** 126	Find the breadth first.
7	**(b)** $3\frac{1}{6}$	Convert to improper fractions or borrow.
8	**(b)** 2	A point is inside if its distance from O is less than the radius.
9	**(c)** 18	Lines of symmetry = $360 \div$ angle.
10	**(c)** -13	Add step by step.
11	**(c)** 18	Number of steps from the first term is $(\text{term} - \text{first}) \div d$; the term number is one more.
12	**(c)** 120	6 degrees per minute, or 30° per number on the dial.
13	**(d)** 90	First digit 1-9 and middle digit 0-9, the last digit is fixed: 9×10.
14	**(c)** 75	Find the difference in symbols then multiply.
15	**(a)** 12	Highest common factor.
16	**(a)** 144	Area becomes 4 times.
17	**(d)** 1	Use the LCM of all denominators.
18	**(d)** 180	Triangle angle sum.
19	**(a)** 300	Multiply.
20	**(c)** 8	Add the opposites.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Rule: 20th term = first term $+(20-1)\times$ difference $= 5 + 19 \times 3$ (1); 20th term $= 62$ (1).	1+1
22	Smaller angle $= 150^\circ$ (obtuse) (1); Larger angle $= 360 - 150 = 210^\circ$ (reflex) (1).	1+1
23	Greatest 97431 (1); Smallest 13479 (1).	1+1
24	Highest 75, lowest 35; Difference $= 40$ (1); Total $= 180$ (1).	1+1
25	$24 = 2 \times 2 \times 2 \times 3$, $36 = 2 \times 2 \times 3 \times 3$ (1); HCF $= 12$, LCM $= 72$ (1).	1+1
26	$l + b = 35$, so breadth $= 11$ cm (1); Area $= 264$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Sticks for 12 triangles $= 3 + 2 \times (12-1) = 25$ (1); For 35 sticks: $3 + 2 \times (m-1) = 35$, so $2 \times (m-1) = 32$ (1); $m - 1 = 16$, so 17 triangles (1).	1+1+1
28	Three-quarter turn $= 270^\circ$ and quarter turn $= 90^\circ$ (1); Net turn $= 270 - 90 = 180^\circ$ (1); Type: straight (1).	1+1+1
29	Step 1: 8172 (1); Step 2: 7443 (1); Digits used: arrange 8172 to get the next number (1).	1+1+1
30	Total $= 130$ (1); Fraction $= \frac{25}{130} = \frac{5}{26}$ (1); Others $= 105$ (1).	1+1+1
31	$180 = 2^2 \times 3^2 \times 5$ (1); Number of factors $= (2+1) \times (2+1) \times (1+1) = 18$ (2).	1+2
	OR: Inner lawn: 26 m by 16 m (1); Inner area 416 m^2 (1); Path area $= 600 - 416 = 184$ m^2 (1).	1+1+1
32	Ploughed =\frac{1}{4}+\frac{1}{3}= $\frac{7}{12}$ (1); Remaining $= 1 - \frac{7}{12}$ (1); $= \frac{5}{12}$ (1).	1+1+1
	OR: Area of square $= 36$ cm^2 (1); One triangle is one quarter: 9 cm^2 (1); Angles at centre: four right angles, 360° (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	10th figure $= 5 + (10 - 1) \times 4 = 41$ (1); More than 91 needs $(91 - 5) \div 4$ more steps (1); Steps needed: 13 (1); Figure number 23 (1); Check by substituting (1).	1+1+1+1+1
34	Hours passed $= 2$, hour hand turns $60°$ (1); Minute hand turns 2 full turns $= 720°$ (1); At 3:00 the hour hand points to 3 ($= 90°$) (1); Smaller angle between the hands $= 90°$ (1); Type: right (1).	1+1+1+1+1
35	The last digit is also 2, and the middle two digits are equal: $2 \times 2 + 2m = 10$ so $m = 3$ (1); Number 2332 (1); Reverse is the same: palindrome (1); Add 11: 2343 (1); Not a palindrome (1).	1+1+1+1+1
36	Mon $= 36$, Tue $= 72$, Wed $= 108$ (1); Thursday $= 36 + 72 = 108$ (1); Total for four days $= 36 + 72 + 108 + 108$ (1); $= 324$ (1); Symbols needed for Thursday: 9 (1).	1+1+1+1+1
	OR: Side $= \mathrm{HCF}(450, 360) = 90$ cm (1); Along the length: 5 tiles (1); Along the width: 4 tiles (1); Total $= 20$ tiles (2).	1+1+1+2
37	Hall in cm: 1000 by 800 (1); Tiles along each side: 20 and 16 (1); Number of tiles $= 320$ (1); Cost $= ₹12800$ (2).	1+1+1+2
	OR: Books: \frac{1}{4}$\times 120 = 30$ (1); Rest $= 90$ (1); Bag: 30 (1); Left $= 60$ (2).	1+1+1+2
38	Draw AB = 4 cm (1); Make right angles at A and B (compass method) (1); Mark AD and BC = 4 cm and join DC (1); Half diagonal OA $= 3$ cm (1); Diagonals meet at $90°$ (1).	1+1+1+1+1
	OR: (a) 5 (1); (b) $72°$ (1); (c) 5 (1); (d) $144°$ is 2 turns so it leaves the pattern unchanged (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–10) | Class 6 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 156	Sum of first n even numbers is $n \times (n+1)$.
2	**(c)** 5	30° per hour.
3	**(d)** 29992	Fill the middle digits with 9.
4	**(a)** 165	Total height times scale.
5	**(b)** 36	Lowest common multiple.
6	**(c)** 4800	Area $\times$ rate.
7	**(a)** 60	Divide by 3, multiply by 5.
8	**(a)** 130	Linear pair.
9	**(c)** A square	Only the square among these has 4.
10	**(b)** 0	Opposites cancel.
11	**(c)** 210	Add 20.
12	**(b)** 160	Full turn is 360°.
13	**(a)** 4	Count each step.
14	**(c)** $\frac{6}{29}$	Fraction $= \frac{30}{145}$ (simplified)
15	**(b)** 1	Digit sum must be a multiple of 3.
16	**(b)** 9	Perimeter of square $= 4\times$ side.
17	**(d)** $\frac{5}{8}$	Fraction left $= 1-$ fraction spent.
18	**(d)** 5	Outer radius minus inner radius.
19	**(c)** 60	The square fits after every multiple of 90°.
20	**(a)** 17	Highest minus lowest.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Convert to improper fractions (1); $3\frac{1}{4}$ (1).	1+1
22	Width $= 9 - 4 = 5$ cm (1); Inner diameter $= 8$ cm; outer diameter $= 18$ cm (1).	1+1
23	Sides $= 360 \div 40 = 9$ (1); Order $= 9$ (1).	1+1
24	Add the first two: 7 (1); Then add (-9): -2 (1).	1+1
25	Terms are 3, 7, ... each 4 more than the last (1); 35 is a term but 36 lies between two terms, so it is not a term (1).	1+1
26	Bisector gives two equal angles (1); $\angle AOC = 2 \times 28 = 56^\circ$, type acute (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Angle $= 360 \div 5 = 72^\circ$ (1); After 3 turns: 216° (1); Turns for full rotation $= 5$ (1).	1+1+1
28	Start at 0; after 3 down: -3; After 5 up: 2 (1); After 4 down: -2 (1); Total floors moved $= 12$ (1).	1+1+1
29	There are 30 odd numbers, sum $= 900$ (1); Even numbers: sum $= 30 \times 31$ (1); $= 930$ (1).	1+1+1
30	Anticlockwise total $= 60 + 70 = 130^\circ$ (1); Clockwise 40° is subtracted (1); Net turn $= 90^\circ$ anticlockwise (right) (1).	1+1+1
31	Largest below: 1221 (1); Smallest above: 1331 (1); Check by reversing (1).	1+1+1
	OR: Total $= 280$ kg (1); Heights: 2, 3, 4, 5 cm (1); Tallest bar: 5 cm (week 4) (1).	1+1+1
32	LCM of 12 and 18 is 36 (1); After 36 minutes (1); Time: 9:36 (1).	1+1+1
	OR: First area $= 80$ cm^2 (1); Second area $= 144$ cm^2 (1); Larger: second, difference $= 64$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	After first jump: 5 (1); After second: 0 (1); After third: -9 (1); Distance from $0 = 9$ (2).	1+1+1+2
34	Sum of first 9 numbers $= \frac{9\times10}{2} = 45$ (1); Sum of cubes $= 2025$ (1); $45^2 = 2025$ (1); They are equal, so the cube sum is a perfect square (2).	1+1+1+2
35	$\angle AOD = 30 + 45 + 60 = 135^\circ$ (1); Type: obtuse (1); Angle left $= 180 - 135 = 45^\circ$ (1); Check with the sum of the parts (2).	1+1+1+2
36	Try arrangements (1); Closest number is 5137 (2); Difference $= 137$ (2).	1+2+2
	OR: Total $= 115$ (1); Maximum: Kabaddi with 60 (1); Fraction $= \frac{60}{115} = \frac{12}{23}$ (1); Scale 1 cm = 10 students: height $= 6$ cm (2).	1+1+1+2
37	LCM of 4, 5, 6 $= 60$ (1); Smallest number $= 61$ (1); Next number $= 121$ (1); Check remainders (2).	1+1+1+2
	OR: Area of big square $= 400$ cm^2 (1); Area of four small squares $= 4 \times 9 = 36$ cm^2 (1); Remaining area $= 364$ cm^2 (1); Perimeter stays the same as the original $= 4 \times 20 = 80$ cm (1); Cutting corner squares does not change the perimeter (1).	1+1+1+1+1
38	Fraction added =\frac{3}{4}−\frac{1}{4}= $\frac{1}{2}$ (1); This part is 20 litres (2); Capacity $= 20 \div \frac{1}{2} = 40$ litres (2).	1+2+2
	OR: Width $= 8 - 5 = 3$ m (1); Outer diameter $= 16$ m (1); Inner diameter $= 10$ m (1); Check with the compass-drawn figure (2).	1+1+1+2

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–10) | Class 6 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 21	Sticks $= 3 + 2 \times (n-1) = 2n+1$.
2	**(b)** 27	A bisector divides the angle into two equal parts.
3	**(b)** 1441	Keep the first half and mirror it; if that is not bigger, increase the middle.
4	**(d)** 65	Add the symbols, then multiply by the value of one symbol.
5	**(c)** 994	Divide 999 by the number.
6	**(a)** 100	Side of square $=$ perimeter $\div 4$.
7	**(d)** $\frac{3}{8}$	Take the average of the two fractions.
8	**(c)** 8	Side $=$ perimeter $\div 4$.
9	**(a)** 10	Add.
10	**(a)** -30	Subtract the descent.
11	**(b)** 625	Consecutive triangular numbers add to a square.
12	**(a)** 270	Add the two turns in degrees.
13	**(d)** 465	Pair the numbers: first with last.
14	**(c)** 0	Count.
15	**(d)** 9	Count them carefully.
16	**(c)** 256	Inner side $= a - 2 \times 2$.
17	**(a)** 17	Whole times denominator plus numerator.
18	**(a)** 24	4 diameters.
19	**(d)** 5	Count the letters with a fold line.
20	**(a)** -7 and 12	Check each pair.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	87, 165, 726, 1353, 4884 (1); Palindrome 4884 (1).	1+1
22	VI-A $= 50$ students; VI-B $= 70$ students (1); Symbols $= 7$ (1).	1+1
23	1234 divided by 9 leaves remainder 1 (1); Add 8 to get 1242 (1).	1+1
24	Area $= 96$ m^2 (1); Cost $= 96 \times 75 =$ ₹7200 (1).	1+1
25	Number $+$\frac{1}{2} of number $=$\frac{3}{2} of number (1); Number $= 18 \div$\frac{3}{2}$= 12$ (1).	1+1
26	$\angle AOD = 180 - 40 = 140^\circ$ (1); $\angle OAB = (180-40) \div 2 = 70^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Mixed to improper: $\frac{5}{2}$ (1); Add $\frac{1}{3}$ (1); Subtract $\frac{1}{4}$: result $2\frac{7}{12}$ (1).	1+1+1
28	Between neighbours: 6 cm (1); First to third: 12 cm (1); Total length $= 6 \times 3 = 18$ cm (1).	1+1+1
29	H: order 2, lines 2; A: order 1, lines 1; T: order 1, lines 1 (1); Compare each pair (1); Counting complete (1).	1+1+1
30	After first: 100; After second: -150 (1); After deposit: -30 (1); Negative means overdrawn (1).	1+1+1
31	$T_{20} = 210$, $T_{21} = 231$ (1); Sum $= 441$ (1); $21^2 = 441$ matches (1).	1+1+1
	OR: Total segments $= \frac{5\times4}{2} = 10$ (1); Segments having A as an end point: 4 (1); Segments without A: $10 - 4 = 6$ (1).	1+1+1
32	Pairs sum to 41 (1); Number of pairs $= 40 \div 2 = 20$ (1); Sum $= 20 \times 41 = 820$ (1).	1+1+1
	OR: Count more than 2: 8 (1); Count exactly 2: 2 (1); Use tally marks to count carefully (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	square: 90° (1); regular octagon: 45° (1); Larger angle: square; difference $= 45^\circ$ (1); Total lines $= 4 + 8 = 12$ (2).	1+1+1+2
34	Coldest to hottest: -8, -5, -2, 3, 11 (1); Hottest 11, coldest -8 (1); Difference $= 19$ (1); Sum $= -1$ (2).	1+1+1+2
35	12th figure $= 3 + (12 - 1) \times 5 = 58$ (1); More than 108 needs $(108 - 3) \div 5$ more steps (1); Steps needed: 11 (1); Figure number 23 (1); Check by substituting (1).	1+1+1+1+1
36	Hours passed $= 5$, hour hand turns 150° (1); Minute hand turns 5 full turns $= 1800^\circ$ (1); At 7:00 the hour hand points to 7 ($= 210^\circ$) (1); Smaller angle between the hands $= 150^\circ$ (1); Type: obtuse (1).	1+1+1+1+1
	OR: The last digit is also 3, and the middle two digits are equal: $2 \times 3 + 2m = 14$ so $m = 4$ (1); Number 3443 (1); Reverse is the same: palindrome (1); Add 11: 3454 (1); Not a palindrome (1).	1+1+1+1+1
37	Mon $= 60$, Tue $= 75$, Wed $= 90$ (1); Thursday $= 60 + 75 = 135$ (1); Total for four days $= 60 + 75 + 90 + 135$ (1); $= 360$ (1); Symbols needed for Thursday: 9 (1).	1+1+1+1+1
	OR: Side $= \text{HCF}(540, 420) = 60$ cm (1); Along the length: 9 tiles (1); Along the width: 7 tiles (1); Total $= 63$ tiles (2).	1+1+1+2
38	Hall in cm: 1200 by 900 (1); Tiles along each side: 30 and 22 (1); Number of tiles $= 660$ (1); Cost $= ₹16500$ (2).	1+1+1+2
	OR: Books: \frac{1}{6}$\times 240 = 40$ (1); Rest $= 200$ (1); Bag: 80 (1); Left $= 120$ (2).	1+1+1+2

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